Production Function Estimation Under Imperfect Competition: Evidence from Czech Construction Procurement

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Research Question. Standard proxy-variable estimators for production functions [1] invert a scalar unobservable (productivity) from input demand, implicitly assuming perfect competition [8, 9]. Ackerberg and De Loecker [2] show that under imperfect competition, input demand also depends on the competitive environment, contaminating productivity and markup estimates. When such markups serve as a dependent variable in a treatment evaluation, the bias is *differential* across treated and control firms—distorting causal estimates. In my MSc thesis [7], I estimate firm-level markups in Czech construction following De Loecker and Warzynski [8] and measure the causal effect of public procurement on markups (ATT ≈15%, declining from 30% to 10% over 2006–2021). This proposal takes that working paper as a base and asks: *how does incorporating imperfect competition—via the ADL sufficient-statistic approach—change the markup distribution and procurement treatment effects?*

Data. I combine two Czech datasets covering 2005–2021: (i) MagnusWeb firm-level financials (CZ-NACE F, 1,297 firms, 7,261 firm-year obs.) with sales, COGS, employees, and fixed assets; and (ii) the Datlab public procurement register with contract values, procedure types, and bidder identities. The procurement register records bidder identities per tender, enabling construction of competitor sets from observed bidder overlap—precisely where the ADL framework [2] generates identification gains. The sample period spans several transparency reforms—the 2012 single-bidding ban, the 2014 bearer share abolition, and the 2017 Register of Contracts [4]—providing exogenous time-series variation in competition intensity.

Model and Approach. The starting point is a translog production function estimated via ACF [1]. The extension proceeds in four steps:

1. *Baseline.* Starting from the ACF translog estimates in [7], compute markups $\mu_{jt} = \hat{\theta}^V_{jt}/\alpha^V_{jt}$ under perfect competition.
2. *Imperfect competition.* Re-estimate conditioning the control function on the competitive environment [2]: total competitor output Q_{-jt} under Cournot, or an inclusive value Φ_{-jt} under differentiated products. Use competitors' fixed inputs and lagged productivity as oligopoly instruments.
3. *Comparison.* Compare markup distributions and test whether the procurement ATT is robust. Because procurement firms face a different competitive environment than non-procurement firms, the standard estimator is differentially biased.
4. *Policy evaluation.* Exploit Czech transparency reforms—the 2012 single-bidding ban and the 2017 Register of Contracts [4]—as exogenous shifters of competition intensity to test whether corrected markups respond differently to institutional changes than standard estimates.

Construction output is heterogeneous, making physical quantities non-comparable. The QEC framework [3] shows deflated revenues are the appropriate output measure, with competitors' deflated revenues as a directly observable sufficient statistic; I further validate using [10].

Contributions. (i) First application of the imperfect-competition PF framework to procurement, where market structure is directly observed. (ii) Quantification of bias from the perfect-competition assumption [2, 6]. (iii) Procurement markets as a natural source of competitive variation aiding identification (cf. [5]), with transparency reforms [4] as exogenous shifters of competition intensity.

Referee Reports. *Methodology:* Ackerberg and De Loecker [2], the theoretical foundation for the correction this project implements. *Application:* Gibbon and Schain [11], the closest published paper to my project—it applies the same DLW/ACF framework to estimate markups.

References

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Data Sources. MagnusWeb (firm financials, `https://www.magnusweb.cz/`), Datlab (procurement register, `https://datlab.eu/`), OECD (industry deflators, `https://stats.oecd.org/`). Replication package: `https://github.com/marek-chadim/Markups-and-Public-Procurement`.